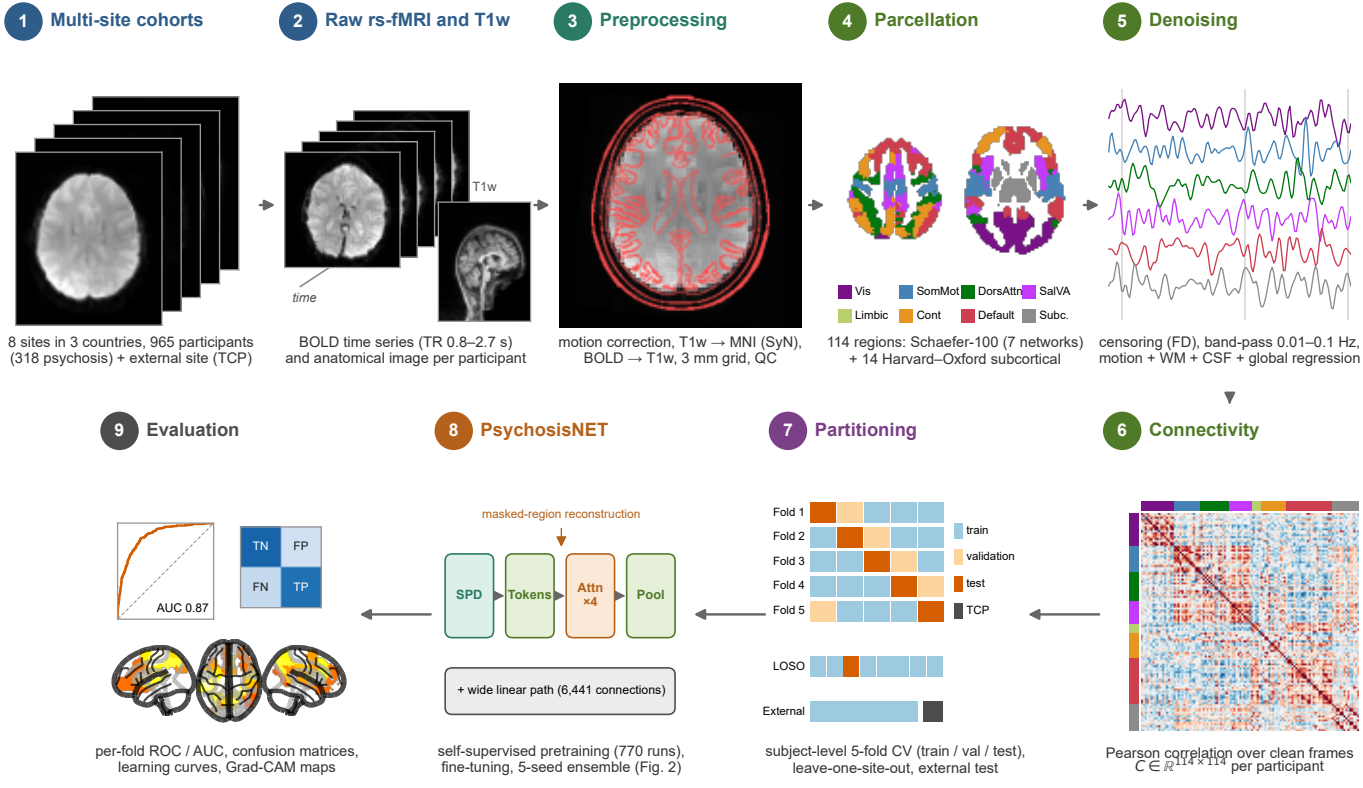

1 Multi-site cohorts
2 Raw rs-fMRI and T1w
3 Preprocessing
4 Parcellation
5 Denoising
T1w
time
Vis
SomMot
DorsAttn
SalVA
Limbic
Cont
Default
Subc.
8 sites in 3 countries, 965 participants (318 psychosis) + external site (TCP)
BOLD time series (TR 0.8–2.7 s) and anatomical image per participant
motion correction, T1w → MNI (SyN), BOLD → T1w, 3 mm grid, QC
114 regions: Schaefer-100 (7 networks) + 14 Harvard–Oxford subcortical
censoring (FD), band-pass 0.01–0.1 Hz, motion + WM + CSF + global regression
9 Evaluation
8 PsychosisNET
7 Partitioning
6 Connectivity
AUC 0.87
TN
FP
FN
TP
masked-region reconstruction
SPD
Tokens
Attn ×4
Pool
+ wide linear path (6,441 connections)
Fold 1
Fold 2
Fold 3
Fold 4
Fold 5
train
validation
test
TCP
LOSO
External
per-fold ROC / AUC, confusion matrices, learning curves, Grad-CAM maps
self-supervised pretraining (770 runs), fine-tuning, 5-seed ensemble (Fig. 2)
subject-level 5-fold CV (train / val / test), leave-one-site-out, external test
Pearson correlation over clean frames
$C \in \mathbb{R}^{114 \times 114}$ per participant